

Yunhao Dong

347-916-2188 | yunhao017@gmail.com | linkedin.com/in/yunhaodong | github.com/yunhaod | yunhaod.github.io/portfolio

EDUCATION

New York University, Tandon School of Engineering

Master of Science in Computer Engineering

Bachelor of Science in Computer Engineering | Dean's List | GPA: 3.852

New York, NY

Expected May 2027

Sep. 2022 - May 2026

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (MySQL), HTML/CSS, C#, Verilog, Swift

Softwares: OnShape, Altium, Wireshark, MATLAB, Cadence

Systems Engineering, Linux, IoT, Hardware Design & Development, RTOS, Oscilloscope, Multimeter, STM32

EXPERIENCE

My Memories LLC

Hardware Engineer

Nov. 2025 – Present

New York, NY

- Developed embedded device using PDM microphone for audio acquisition & Bluetooth LE transmission
- Built an iOS app that reconstructs BLE audio data into WAV files with highly intelligible (95%) speech

NYU Tandon Computer Science & Engineering Department

Course Assistant

Sep. 2024 – Present

Brooklyn, NY

- Conducted office hours weekly to assist 100+ students to clarify concepts, course material, and coding assignments
- Facilitated lab sessions on Python to guide 30 students through hands-on exercises to ensure topic comprehension

Honeywell International, Building Automation Team

Application Engineer Intern

June 2025 – Aug. 2025

Northford, CT

- Researched and tested Bluetooth LE channel sounding with Nordic nRF54L15 and Silicon Labs boards, achieving 80% accuracy in indoor positioning
- Designed an asset tracking system prototype for hotel tray retrieval using beacon technology, collaborating with cross-functional engineers to evaluate power consumption and optimize battery life constraints
- Traced Modbus RTU messages with custom VBA program to analyze device behavior using a protocol converter

NYU Vertically Integrated Projects: Robotics Design Team

Systems Engineer

Jan. 2024 – May 2025

Brooklyn, NY

- Programmed C++ control logic for manual & autonomous robot operations in NASA's Lunabotics Competition
- Optimized I2C communication in Linux between NVIDIA Jetson Orin Nano with Teensy, reducing latency by 30%
- Interfaced Raspberry Pi4 with Teensy through I2C for data and command transfer, enabling wireless control

NYU Undergraduate Summer Research

Student Researcher

June 2023 – Aug. 2023

Brooklyn, NY

- Integrated C# scripts for game objects, UDP protocol, and Python computer vision to track hand movements across webcam with 90% accuracy in Unity
- Designed a wristband PCB in Altium to gather data from hand gestures, reducing prototype size by 40%

PROJECTS

MistLink | Matter, ESP32, C++, Arduino

Oct. 2024 – Dec. 2024

- Prototyped and implemented a humidifier with ESP32, integrated with the Matter protocol for IoT communication
- Troubleshoot firmware for interoperability between humidifier, Google Nest Hub, and Google Home app

PetPlatter | Websockets, AWS EC2, Swift

July 2024 – Aug. 2024

- Launched a WebSocket server on AWS EC2 virtual machine to facilitate real-time communication to control a pet feeding device, establishing low-latency interaction between application and hardware
- Implemented aa swift app to control device for feeding food and configured ESP32 to handle incoming WebSocket messages to execute command

SecureMotion | MQTT, ESP32, C++, iOS

June 2024 – July 2024

- Developed iOS app with swift for real-time home security monitoring and control, utilizing MQTT for bidirectional communication and ESP32 microcontroller for feedback
- Enhanced IoT communication protocol by integrating an authentication mechanism to ensure data privacy
- Streamlined message handling for effective data transmission, resulting in improved system performance by 20%